

Tutorial 3: Solutions

Linear Transformations

INF1004 Mathematics II

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Solution 1: Cramer's Rule (Part 1)

Linear System (Currents i_1, i_2, i_3):

$$\begin{cases} 1i_1 + 1i_2 + 1i_3 = 6 \\ 1i_1 - 1i_2 + 1i_3 = 0 \\ 2i_1 + 1i_2 - 1i_3 = 6 \end{cases}$$

Step 1: Calculate the Main Determinant (D)

Expand along the first row:

$$D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{vmatrix} = 1 \begin{vmatrix} -1 & 1 \\ 1 & -1 \end{vmatrix} - 1 \begin{vmatrix} 1 & 1 \\ 2 & -1 \end{vmatrix} + 1 \begin{vmatrix} 1 & -1 \\ 2 & 1 \end{vmatrix}$$

$$D = 1(1 - 1) - 1(-1 - 2) + 1(1 - (-2)) = 0 + 3 + 3 = 6$$

Since $D \neq 0$, the system has a unique solution.

Solution 1: Cramer's Rule (Part 2)

Step 2: Calculate D_{i_n}

$$D_{i_1} = \begin{vmatrix} \mathbf{6} & 1 & 1 \\ \mathbf{0} & -1 & 1 \\ \mathbf{6} & 1 & -1 \end{vmatrix} = 12$$

$$D_{i_2} = \begin{vmatrix} 1 & \mathbf{6} & 1 \\ 1 & \mathbf{0} & 1 \\ 2 & \mathbf{6} & -1 \end{vmatrix} = 18$$

$$D_{i_3} = \begin{vmatrix} 1 & 1 & \mathbf{6} \\ 1 & -1 & \mathbf{0} \\ 2 & 1 & \mathbf{6} \end{vmatrix} = 6$$

Step 3: Final Values

$$i_1 = \frac{D_{i_1}}{D} = \frac{12}{6} = 2$$

$$i_2 = \frac{D_{i_2}}{D} = \frac{18}{6} = 3$$

$$i_3 = \frac{D_{i_3}}{D} = \frac{6}{6} = 1$$

Solution Vector

$$\mathbf{i} = [2 \quad 3 \quad 1]^T$$

Solution 2: Gaussian Elimination

Step 1: Forward Elimination

Augmented Matrix:
$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 2 & 3 & 6 \\ 3 & -1 & -1 & 2 \end{array} \right]$$

- $R_2 \rightarrow R_2 - 2R_1: [0, 0, 1 \mid 2]$
- $R_3 \rightarrow R_3 - 3R_1: [0, -4, -4 \mid -4]$

Swap R_2 and R_3 :

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & -4 & -4 & -4 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

For Row Echelon Form, $R_2 = R_2 / -4$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

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For Row Echelon Form, $R_2 = R_2 / -4$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

Step 2: Back Substitution

- From R3: $c = 2$
- From R2: $b + c = 1 \implies b + (2) = 1 \implies b = -1$
- From R1:
 $a + b + c = 2 \implies +(-1) + 2 = 2 \implies a + 1 = 2 \implies a = 1$

Solution 3: Consistency Check

① $\left[\begin{array}{cc|c} 1 & 1 & 3 \\ 2 & -1 & 0 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 1 & 3 \\ 0 & -3 & -6 \end{array} \right] \implies -3y = -6 \implies y = 2 \implies x + y = 3 \implies x = 1. \text{ Consistent.}$

② $\left[\begin{array}{cc|c} 1 & -2 & 4 \\ 2 & -4 & 10 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & -2 & 4 \\ 0 & 0 & 2 \end{array} \right]. \text{ Last row implies } 0 = 2,$
which shows inconsistency, no solution.

③ $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 2 & 2 & 2 & 10 \\ 1 & -1 & 1 & 0 \end{array} \right] \xrightarrow{R_3 \leftrightarrow R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 1 & -1 & 1 & 0 \\ 2 & 2 & 2 & 10 \end{array} \right] \xrightarrow{\begin{array}{l} R_3 \rightarrow R_3 - 2R_1 \\ R_2 \rightarrow R_2 - R_1 \end{array}} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 5 \\ 0 & -2 & 0 & -5 \\ 0 & 0 & 0 & 0 \end{array} \right]. \text{ Last row implies } 0 = 0, \text{ other rows are also consistent, so equation is consistent with infinite solutions.}$

Solution 4a: Inverse of Matrix A

Given $A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix}$

1. **Determinant:** $\det(A) = 1 \begin{vmatrix} 1 & 0 \\ 0 & 2 \end{vmatrix} - 0 + 1 \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = 1(2) + 1(-1) = 1 \neq 0$.

2. **Cofactor Matrix:** Calculated via minors with alternating signs:

$$\text{Cof}(A) = \begin{pmatrix} 2 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

3. **Adjoint and Inverse:** Since $\text{adj}(A) = \text{Cof}(A)^T$, and $\text{cof}(A)$ is symmetric

here, $\text{adj}(A) = \begin{pmatrix} 2 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$.

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A) = \begin{pmatrix} 2 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

Solution 4b: Inverse of Matrix B

Given $B = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 7 & 8 & 9 \end{pmatrix}$

1. Observation: Notice that the second row (R_2) is exactly twice the first row (R_1):

$$R_2 = 2 \cdot R_1 \implies [2, 4, 6] = 2 \cdot [1, 2, 3]$$

2. Determinant Properties: If one row is a scalar multiple of another, the rows are linearly dependent. By the properties of determinants:

$$\det(B) = 0$$

Conclusion: Since the determinant is zero, Matrix B is singular. An inverse does not exist (B^{-1} is undefined).

Solution 4c: Inverse of Matrix C

Given $C = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

1. Determinant: $\det(C) = 2 \times 1 \times 1 = 2$.

2. Adjoint Matrix: Find the cofactor matrix $\text{cof}(C) = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ and

transpose it: $\text{adj}(C) = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$

3. Final Inverse:

$$C^{-1} = \frac{1}{2} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} = \begin{pmatrix} 0.5 & -0.5 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Solution 5: MMORPG Application

Let x = Mythril Helm, y = Dragon Plate, z = Speed Boots.

System of Equations:

- 1 Total Items: $x + y + z = 20$
- 2 Total Attack: $5x + 20y + 2z = 235$
- 3 Total Defense: $15x + 25y + 5z = 350$

Solving (simplified): Use any of the methods (Gaussian elimination/Cramer's Rule/Inverse calculation)

Final Loadout

$$x = 5 \text{ (Helms)}, \quad y = 10 \text{ (Plates)}, \quad z = 5 \text{ (Boots)}$$

Augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 20 \\ 5 & 20 & 2 & 235 \\ 15 & 25 & 5 & 350 \end{array} \right]$$

Row operations (skipping obvious arithmetic):

$$R_2 \leftarrow R_2 - 5R_1, \quad R_3 \leftarrow R_3 - 15R_1$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 20 \\ 0 & 15 & -3 & 135 \\ 0 & 10 & -10 & 50 \end{array} \right]$$

Normalize R_2 and eliminate below:

$$R_2 \leftarrow \frac{1}{15} R_2, \quad R_3 \leftarrow R_3 - 10R_2$$

$$\Rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 20 \\ 0 & 1 & -\frac{1}{5} & 9 \\ 0 & 0 & -8 & -40 \end{array} \right]$$

Back-substitution gives:

$$z = 5, \quad y - 1 = 9 \Rightarrow y = 10, \quad x + 10 + 5 = 20 \Rightarrow x = 5.$$

$$\boxed{x = 5, \quad y = 10, \quad z = 5}$$

Solutions: Non-Commutativity of 3D Rotations

1. Individual Matrices (90° rotations)

Given $\theta = 90^\circ$ and $\phi = 90^\circ$, we have $\cos(90^\circ) = 0$ and $\sin(90^\circ) = 1$.

Substituting these values into the definitions:

- **Rotation about x-axis (A):**

$$A = R_x(90^\circ) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$$

- **Rotation about y-axis (B):**

$$B = R_y(90^\circ) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}$$

2. Order 1: x-axis then y-axis (M_1)

The transformation is $M_1 = R_y(90^\circ)R_x(90^\circ) = B \cdot A$.

$$M_1 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ -1 & 0 & 0 \end{pmatrix}$$

3. Order 2: y-axis then x-axis (M_2)

The transformation is $M_2 = R_x(90^\circ)R_y(90^\circ) = A \cdot B$.

$$M_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

Conclusion: Comparing the results, $M_1 \neq M_2$. This confirms that 3D rotations are **non-commutative**¹; the order in which rotations are applied affects the final orientation of the object.

Interesting discussion with a cool dice example: <https://tinyurl.com/btw3u69j>

¹Note that 2D rotations are a special case when matrix multiplication is commutative, and order of rotation about x and y axis doesn't matter.